

REISHAUER

Gear Grinding Technology

Reishauer Clamping Device

DH

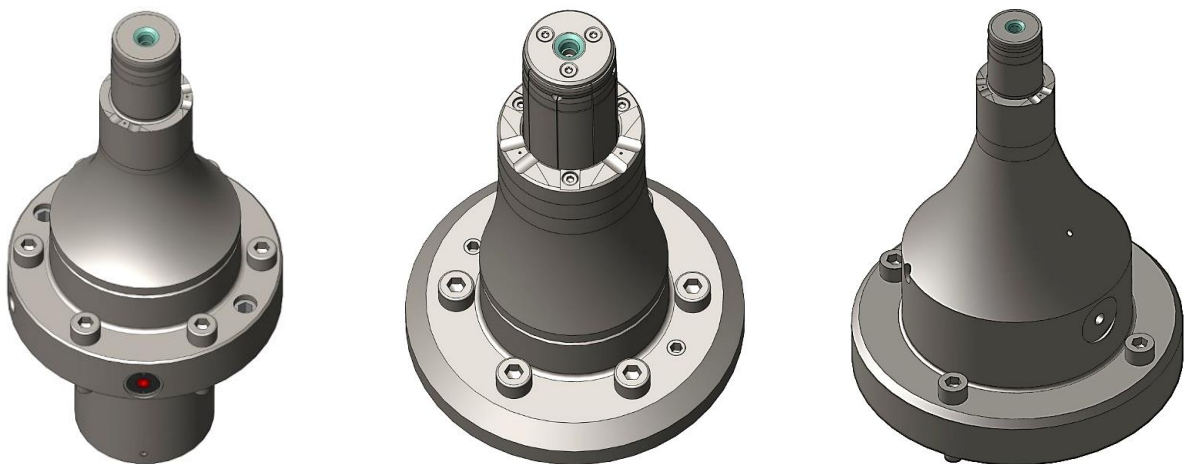
Clamping arbor, hydraulic

Installation and Operating Manual

Item number: 60051100 - DH-BA1

Doc. No.: 60051122

Original instructions





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1 General information

1.1 On these instructions

These instructions facilitate safe and efficient machine handling. These instructions form part of the machine and must be kept accessible to all personnel in the vicinity of the machine.

Carefully read these instructions before beginning any work.

A fundamental requirement for safe working is adherence to all safety and handling instructions in these operating instructions.

Keep these instructions for future reference. If the machine is passed on to third parties, also provide them with these instructions.

1.2 Key to symbols

The safety instructions are divided into three categories. They have the following meaning:

WARNING



Information about a potentially dangerous situation, which can result in death or serious bodily injuries if not avoided.

CAUTION



Information about a potentially dangerous situation, which can lead to minor injuries if not avoided.

ATTENTION

Information about a potentially dangerous situation, which can lead to damage to property and the environment if not avoided.

Certain instructions in this manual are specially identified in order to ensure quick access to the information.



Information for handling and use, as well as technical instructions.
Non-compliance may lead to avoidable additional cost.

1.3 Customer service

Our customer service is available for technical enquiries. See the last page of these instructions for contact details.

2 For your safety

This section outlines the safety procedures that must be strictly observed when operating the device.

2.1 Approved use

The Reishauer Clamping Device **DH** is intended for industrial use within a machine tool.

It is used only for the mounting, tensioning and secure fastening of customer-specific workpieces, which are gripped using a bore. The workpieces must have the connection dimensions specified in the technical data. The clamping fixture must be installed in a vertical position.

If the clamping fixture is installed under rotation within another machine other than the Reishauer gear grinding machine, this machine must meet all the essential requirements of the EC Machinery Directive.

Intended use also includes the observation of all information in these instructions.

Any use not falling within the scope of intended use is considered misuse.

WARNING



Danger if misused!

Misuse of the clamping fixture may result in hazardous situations. In particular, the device must not be used for the following:

- > Tensioning workpieces, materials, and/or tools other than those mentioned above.
- > Commencing the clamping bush procedure without the workpiece or protection sleeve in place.
- > Operating the clamping fixture outside the values specified in the "Technical data" section.
- > Retrofitting, converting, or modifying the construction or individual equipment parts with the aim of changing the device's area of application or usage.

The manufacturer shall not be held liable for any damage caused as a result of misuse of the equipment.

2.2 Basic hazards

Residual risks are identified in the following section. These risks can also occur when the device is used as intended.

In order to reduce risk of personal injury and material damage and to prevent hazardous situations, please observe the safety instructions listed here and throughout the document.

Hazard areas

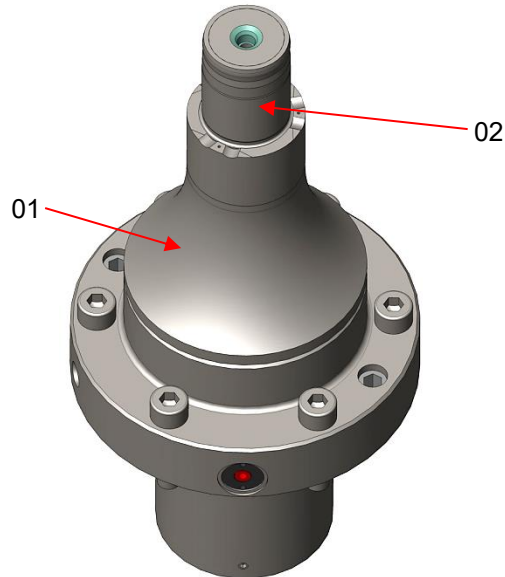


Fig. 1: Hazard areas

- Entire clamping fixture (01) subject to hydraulic energy.
- Clamping bush (02) subject to tensioning without a workpiece or protection sleeve.
- Clamping bush (02) if clamping pressure incorrectly set.
- Crushing points between the clamping fixture, workpiece and counter support during axial positioning of workpieces by the operator → *Working in Manual Mode with Open Safety Doors*.

Entire clamping fixture subject to hydraulic energy

WARNING



Risk of injury from hydraulic energy!

The clamping fixture is under hydraulic pressure (pressure in the system). Hydraulic energy can cause serious injury or death.

- > Never attempt to repair or open the devices without authorization.
- > Always send the devices to the manufacturer for repair.

Breakage of the clamping bush**WARNING**

Risk of injury and material damage due to improper tensioning of the clamping bush!

Tensioning the clamping bush without a workpiece or protection sleeve can result in destruction of the clamping bush and serious injuries.

- > Never tension the clamping bush without introducing a workpiece or protection sleeve.
- > Ensure that surface to be clamped is fully cylindrical and does not have any interruptions (i.e. grooves, bores, threads etc.).

Incorrectly Set Clamping Pressure**WARNING**

Risk of injury and material damage due to incorrectly set clamping pressure!

A clamping pressure that is too high can destroy the clamping fixture and cause serious injury. If the clamping pressure is set too low, the workpiece will not be sufficiently tensioned. This can lead to a collision with machine parts. Parts may be ejected, and the pressure medium may escape at high pressure. This can lead to serious or fatal injury as well as material damage.

- > Always set the clamping pressure at the clamping fixture according to the appropriate specification.
- > Never change the clamping pressure without authorization.
- > Ensure that the correct clamping pressure is set every time the clamping fixture is changed.

Working with open protection doors in manual mode**WARNING**

Danger of injury due to moving machine components with open protection doors!

A series of setup tasks are performed with the protection doors open. Necessary machine motions are rendered safe by the confirmation function. Even so, there is still a danger of injury in principle.

- > Perform the tasks with the protection doors open with extreme care and caution.
- > Ensure that tasks with open protection doors are only performed by authorized operating staff.
- > Do not reach into or linger near moving components.
- > The machine may only be operated by one person at a time.

2.3 Personnel requirements

WARNING



Danger of injury as a result of insufficient personnel training!

Insufficiently qualified personnel do not recognise the hazards in the working zone. This can lead to serious injuries.

- > All activities should only be performed by suitably qualified personnel.
- > Keep unauthorised and insufficiently qualified personnel out of the working zone.

The following qualifications are specified in the instructions:

- **Skilled personnel**
are, by reason of their technical training, knowledge and experience, as well as their knowledge of the relevant regulations, in a position to carry out the work entrusted to them and able to recognise and avoid possible dangers on their own initiative.
- **Manufacturer**
Certain tasks may only be carried out by the manufacturer's skilled personnel. Any other personnel are not authorised to carry out work. Please contact our customer service when any work is required.

Only persons expected to perform work reliably are authorised as personnel. Persons whose reactions are impaired, e.g. by drugs, alcohol or medicines, are not authorised.

2.4 Personal protective equipment

Always wear the following when working:



Protective workwear

is close-fitting workwear with low tensile strength, tight sleeves, and no loose-hanging materials.



Safety shoes

provide protection against heavy falling parts and from slipping on slippery surfaces.



Safety goggles

protect eyes from flying parts and liquid splashes.



Chemical-resistant protective gloves

protect hands against aggressive chemicals.

2.5 Responsibility of operating company

The machine is used in an industrial environment. Therefore, the operating company is subject to statutory health & safety obligations.

In particular, the following points apply:

- The operating company must perform a risk assessment for the workplaces in his area of responsibility where the machine is used and issue operating instructions derived from this assessment. These must correspond to current standards and, if necessary, be continuously updated.
- The operating company must ensure that all employees using the machine have read and understood these instructions.
- The operating company is responsible for ensuring that all work carried out on the machine is performed only by employees trained for the specific tasks.
- The operating company must provide personnel with the appropriate personal protective equipment and ensure that the necessary personal protective equipment is worn.

3 Technical data

3.1 Marking

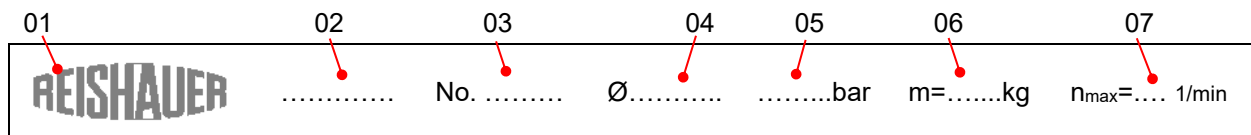


Fig. 2: Marking

01	Manufacturer	05	Clamping pressure
02	Article number	06	Weight (option)
03	Serial number	07	Operating speed
04	Clamping diameter		

3.2 General information

Specification	Value
Locating diameter (connection to work-piece spindle)	See drawing and marking on the clamping fixture
Clamping diameter (connection to work-piece)	
Clamping stroke, max.	See drawing of clamping fixture



Refer to the Appendix for the dimensions in the drawing provided.

3.3 Operating conditions

Specification	Value
Maximum period of continuous operation	-
Speed, max.	See drawing and marking on the clamping fixture
Temperature range	15–40 °C
Relative humidity	15–70%
Fluctuation of the ambient temperature per hour, max.	2°C/h
Hydraulic power ratings	Value
Hydraulic clamping pressure, max.	See drawing and marking on the clamping fixture
Pneumatic power ratings	Value
Pneumatic operating pressure, max.	See separate documents for the grinding machine

4 Construction and function

4.1 Clamping arbors: Overview

The figures below show the typical **DH** clamping arbors, which are fitted using fixing screws (01) either directly to the machine tool workpiece spindle or to a Base Unit. The options are identified by the * symbol.

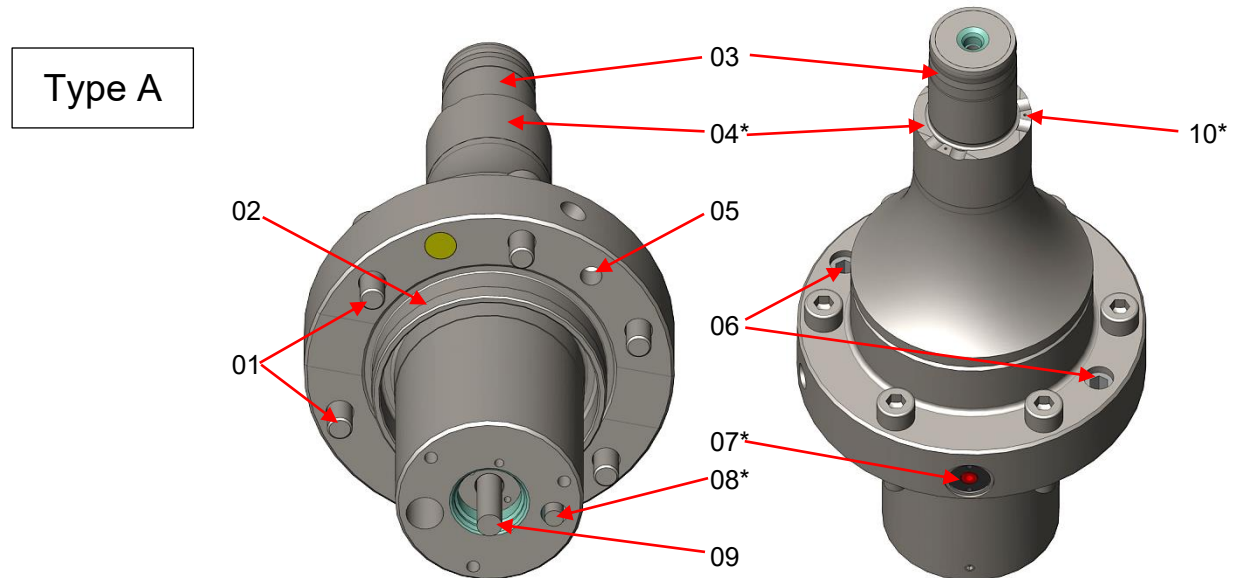


Fig. 3: DH Reishauer Clamping Device with $\varnothing 75$ mm bevel connection for fitting to the workpiece spindle (type A)

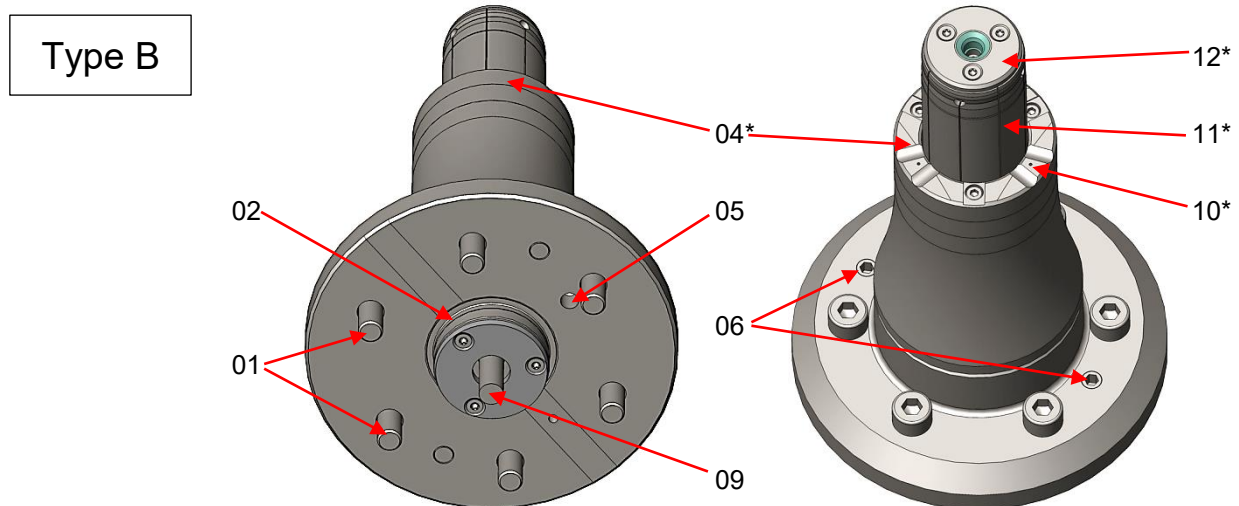


Fig. 4: DH Reishauer Clamping Device with cylindrical pin or bore for fitting to the Base Unit (type B).

Type C

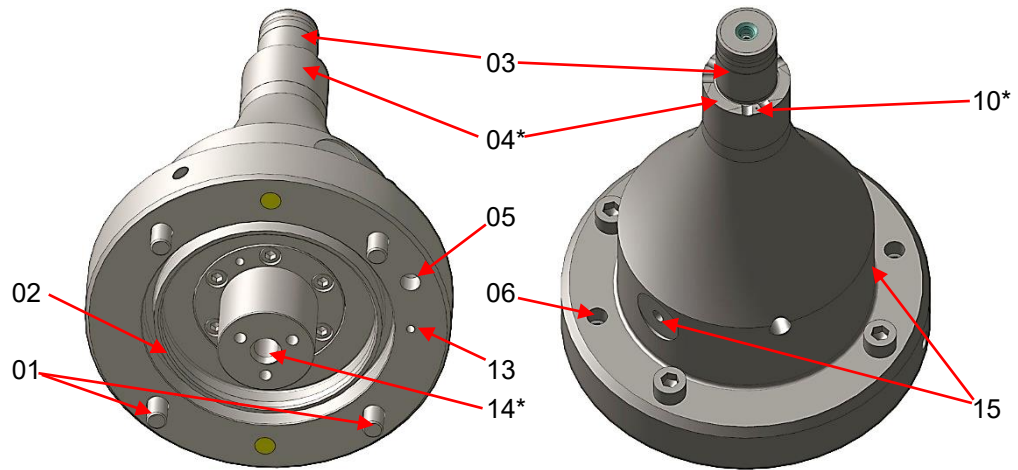


Fig. 5: DH Reishauer Clamping Device with $\varnothing 106$ mm bevel connection for fitting to the workpiece spindle (type C)

01	Fixing screws – connection to machine spindle or Base Unit	09	Operating piston
02	Centering to workpiece spindle or Base Unit	10*	Air holes for part present monitoring (option)
03	Clamping bush or expansion area – mounting for workpiece	11*	Intermediate bush (option)
04*	Support ring (option)	12*	Retaining washer for intermediate bush (option)
05	Bore for positioning pin	13	Hydraulic connection – oil inlet for pressure intensifier
06	Threaded lift-off pin or threaded lift-off pins	14*	Air connection – air connection for e.g. part present monitoring (option)
07*	Clamping stroke control (option)	15	Lifting thread (2x)
08*	Release pin for clamping stroke control (option)		

4.2 Brief description

The DH Reishauer Clamping Devices (hereafter referred to as “clamping fixtures”) are used exclusively for the mounting and tensioning of workpieces using a hydraulically expanding clamping bush (03).

The clamping fixtures are inserted either directly into the machine tool workpiece spindle (types A and C) or into a Base Unit (type B), and centered there in the taper mounting or using a cylindrical pin. The expanding clamping arbors are secured to the workpiece spindle or Base Unit using fixing screws (01).

Using the pressure from the machine, an operating piston (09) located either in a Base Unit or in the clamping fixture itself is actuated and the pressure medium is fed into the clamping bush (03). The clamping bush expands under this pressure and clamps the workpiece.

When tensioning without a workpiece and with high pressure, the clamping fixture may be destroyed due to lack of mechanical counter-pressure.

4.3 Description of the components

4.3.1 Support ring, Pendulum systems (options)

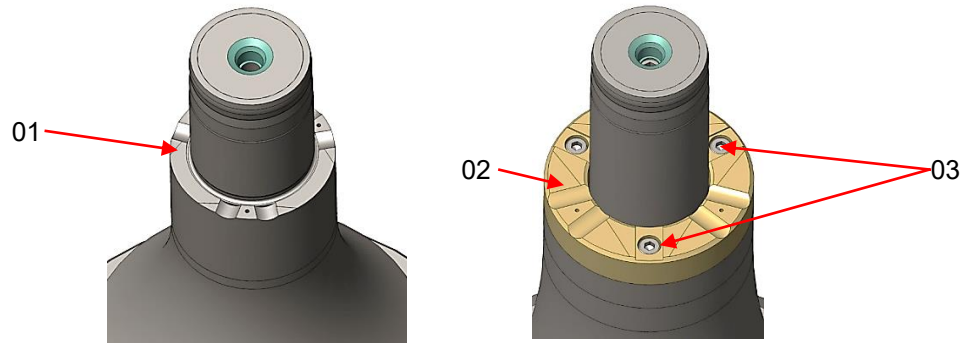


Fig. 6: Integrated support ring;

Replaceable support ring

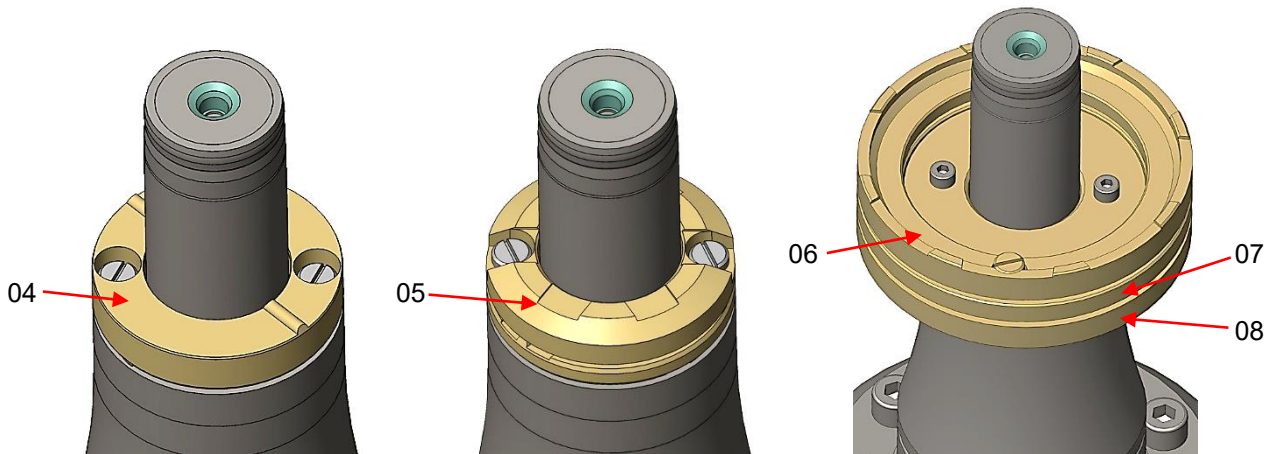


Fig. 7: Pendulum systems:

One-piece

two-piece,

three-piece

01	Integrated support ring	05	Two-piece pendulum system
02	Replaceable support ring	06	Three-piece pendulum system
03	Fixing screws	07	Pendulum bob
04	One-piece pendulum system	08	Pendulum system support ring

The support ring is used for axial positioning of the workpiece. The support ring can be integrated (01) into the clamping fixture or replaceable (02), and can be oscillating (04, 05, 06) or sprung.

Depending on the workpiece, various support ring design options are possible, since the support ring is adjusted to the workpiece and its requirements. Depending on design, the support ring or pendulum system is fixed with a number of fixing screws.

4.3.2 Threaded lift-off pin or threaded lift-off pins

The threaded lift-off pins are used to separate the clamping fixture from the Base Unit or workpiece spindle of the machine.

To separate the clamping fixture, the threaded lift-off pins must be turned clockwise.

4.3.3 Clamping stroke control (option)

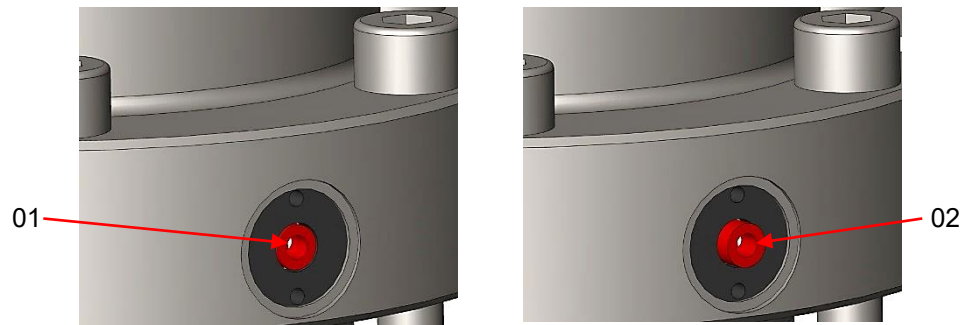


Fig. 8: Control pin of the clamping stroke monitoring

01	Control pin retracted – normal operation	02	Control pin extended – clamping device is not fully functional
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The functionality of the clamping fixture can be checked using the visual clamping stroke control, provided that the function of the clamping stroke control is activated in the machine control.

If the fitted clamping fixture is no longer sufficiently functional due to loss of oil, the red control pin (02) protrudes by approx. 2 mm. At the same time, air escapes from the vent hole in the control pin. In this case, the clamping fixture must be replaced and sent for repair.

4.3.4 Operating piston

The required oil pressure for tensioning is built up in the clamping fixture using the operating piston.

4.3.5 Part present monitoring (option)

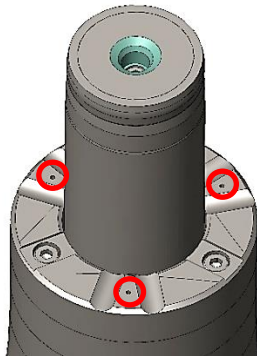


Fig. 9: Part present monitoring

Using compressed air, the part present monitoring checks whether a workpiece is sitting correctly on the clamping fixture. For this purpose, the clamping fixture is provided with air holes (marked in red) on the locating surface. If the workpiece is not present or is not sitting correctly, the air can escape through these air holes. A pressure switch on the machine monitors the dynamic pressure in the clamping fixture.



For further information about setting up the pressure switch, see *Commissioning and Maintenance Instructions* for grinding machine, section *Adjusting the Part Present Monitoring Air Gap Sensor*.

4.3.6 Intermediate bush (option)

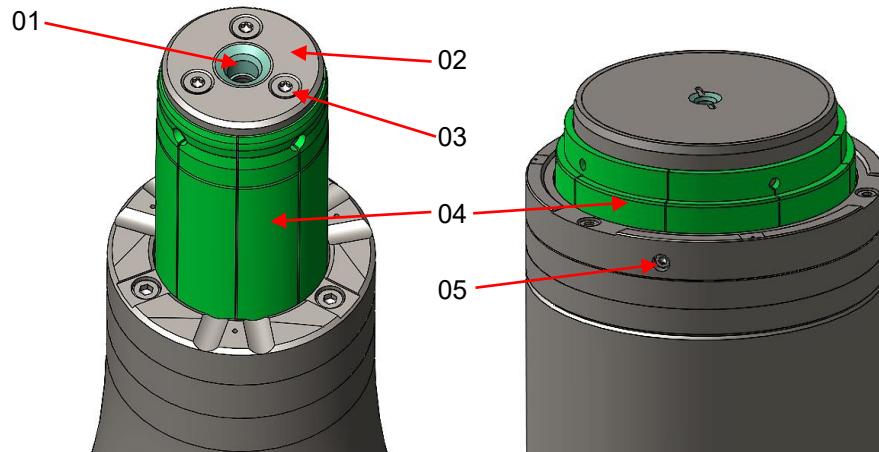


Fig. 10: Intermediate bushes (examples)

01	Pins for intermediate bush holder	04	Intermediate bush
02	Retaining washer for intermediate bush	05	Retaining screw for intermediate bush
03	Fixing screws for retaining washer		

The intermediate bush (04) is used to bridge different hole diameters. Using different intermediate bushes, various workpieces can be tensioned with a single clamping fixture.

The intermediate bush holder can be designed in different ways depending on the size of the clamping fixture. This prevents the intermediate bush from being pulled off axially when unloading the workpiece.

4.3.7 Hydraulic connection – oil inlet for pressure intensifier

The hydraulic connection is used to actuate the internal piston and in so doing to tension and release workpieces at the clamping fixtures. Depending on additional options, there may be a second hydraulic connection or oil line.

4.3.8 Lifting thread

Heavy clamping fixtures (15 kg or more) are provided with lifting threads. The lifting threads are used for lifting the clamping fixture with the customer's own lifting device.

4.4 Connections



For information about the connections, please refer to the drawings provided and the *Construction and Function* section.

5 Commissioning

5.1 Commissioning safety

Personnel

- Technical personnel

Personal protective equipment

- Protective workwear
- Safety shoes

WARNING



Risk of injury due to improper commissioning!

Improper commissioning can lead to serious injury. Before assembling the device, ensure that the machine in which the device is operated, meets the following requirements:

- > The applicable Type C standards for the conception and construction of the relevant machine have been adhered to.
- > The machine in which the device is installed meets all the essential requirements of the EC Machinery Directives.

WARNING



Risk of injury and material damage due to improper installation or removal!

Improper installation or removal can lead to serious injury or material damage and impair the functionality of the clamping system.

- > Perform all operating steps in accordance with the information provided in these instructions.
- > Maintain order and cleanliness in the work area. Parts and tools that are not stacked securely or are lying around are a potential cause of accidents.
- > Do not loosen any additional screws on the clamping system unless expressly instructed to do so in these instructions.

5.2 Preparing the machine tool

1. Provide access to the workpiece spindle of the relevant machine tool or open the movable protective covers → see separate documents for the relevant machine tool.

WARNING



Risk of injury due to moving parts of the machine tool!

Risk of injury due to hydraulic and pneumatic energy!

- > If the machine tool cannot be switched off as described below because the control is required to change the clamping device, ensure that the operating buttons cannot be pressed either unintentionally or without authorization.

2. Switch off the machine tool and secure against reactivation → separate documents for the relevant machine tool.

3. Depressurize the hydraulic system → separate documents for the relevant machine tool.
 4. Ensure that the hydraulic system of the machine tool is switched off and depressurized.
- > The machine tool is prepared.

5.3 Fit clamping fixture to workpiece spindle

5.3.1 Fit type A clamping fixture

Tools

- Lint-free cloth
- Allen wrench set
- Torque wrench with hexagon socket (size 6 mm)

Preparations

1. Prepare machine tool → section 5.2.

Preparing the locating surfaces

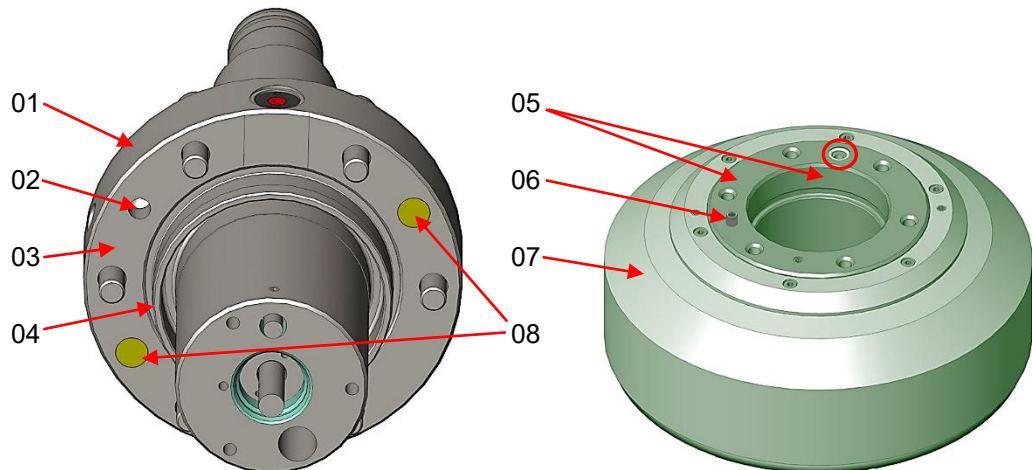


Fig. 11: Position the type A clamping fixture on the workpiece spindle.

2. Ensure that the threaded lift-off pins (08) do not protrude. Adjust if necessary.
3. Clean the plane surface (03) and cone (04) of the clamping fixture (01) with a lint-free cloth and check for damage.



If the locating surfaces are damaged, replace the clamping fixture.

4. Clean the locating surfaces (05) and workpiece spindle (07) of the machine with a lint-free cloth and check for damage.
5. Ensure that the O-ring (marked in red) is present in the workpiece spindle countersink. Insert O-ring if required.

Positioning and securing the clamping fixture

6. Carefully position the clamping fixture (01) on the spindle nose of the work-piece spindle (07), ensuring that it is aligned correctly. Observe the position of the positioning bore (02) for the positioning pin (06), identified by the  symbol.
The positioning pin (06) must be located in the positioning bore (02) provided.

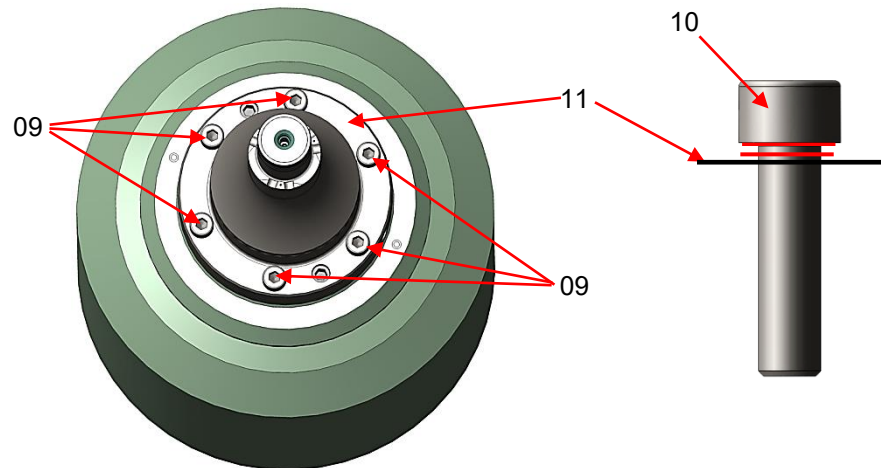


Fig. 12: Secure type A clamping fixture (top view)

09	Fixing screws – hexagon socket cylinder head screws M8 with 6 mm width across flats	11	Clamping fixture
10	Screw head		

7. Position the fixing screws (09) in the bores provided.



The fixing screws must be tightened in the order described below (three steps), otherwise the radial runout will be incorrect.

Turn the fastening screws until they are finger-tight

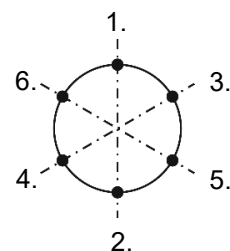


Fig. 13: Turn the screws until they are finger-tight (example)

8. Using only fingers (see figure), tighten fixing screws (09) in a crosswise manner and a clockwise direction, until there is slight resistance.
- > Once the first contact between the screw head (10) and the clamping fixture (11) has been made, stop turning and start work on the next screw.



If pressure is applied while screwing in the fixing screws, the clamping fixture will be inserted into its mounting at an angle. In this case, loosen all the fixing screws, pull the clamping device outwards and upwards until it separates from its mounting, and repeat step 8.

Turn the fastening screws until they are hand-tight

9. Tighten fixing screws (09) manually using a hexagon socket screw key in a crosswise manner.

Tighten the fastening screws to a torque of

10. Tighten the fixing screws (09) in a crosswise manner using an Allen wrench to a torque of max. 20 Nm for M8 screws and 30 Nm for M10 screws.

Check the radial and axial runout

11. Check the radial and axial runout using a gauge with an indicating caliper. Refer to the drawings provided for the clamping fixture for the test centers and radial and axial runout deviations.



If the radial and axial runout deviations suggest a higher value than specified, remove the clamping fixture and repeat steps 2 to 11.

If the deviations are still not permissible, send the clamping fixture to the manufacturer (see section *Return Transportation to Manufacturer*) and fit a new clamping fixture.

12. Fit support ring and intermediate bush if required → *Fit/Remove Support Ring and Pendulum System* and *Fit/Remove Intermediate Bush* sections.
- > The type A clamping fixture is now fitted.

5.3.2 Fit type B clamping fixture

The type B clamping fixture is fitted to a Base Unit.

Preparations

1. Prepare machine tool → section 5.2.
2. Fit Base Unit → instructions provided for the relevant Base Unit.
3. Fit the type B clamping fixture to the Base Unit → instructions provided for the relevant Base Unit.
4. Fit support ring and intermediate bush if required → *Fit/Remove Support Ring and Pendulum System* and *Fit/Remove Intermediate Bush* sections.

5.3.3 Fit type C clamping fixture

Tools

- Lint-free cloth
- Allen wrench set
- Torque wrench with hexagon socket (size 8 mm)

Preparations

1. Prepare machine tool → section 5.2.

Transporting the clamping fixture

2. If required, screw the eye bolts into two lifting threads of the clamping fixture.
3. If required, attach cables or straps with sufficient load capacity to a crane and carefully transport the clamping fixture to the machine tool.

Preparing the locating surfaces

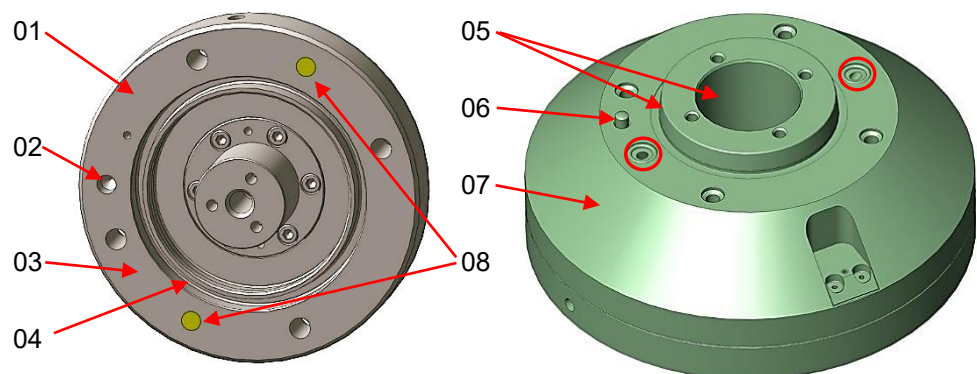


Fig. 14: Position the type C clamping fixture on the workpiece spindle.

4. Ensure that the threaded lift-off pins (08) do not protrude. Adjust if necessary.
5. Clean the plane surface (03) and cone (04) of the clamping fixture (01) with a lint-free cloth and check for damage.



If the locating surfaces are damaged, replace the clamping fixture.

6. Clean the locating surfaces (05) and workpiece spindle (07) of the machine with a lint-free cloth and check for damage.
7. Ensure that the O-rings (marked in red) are present in the workpiece spindle countersinks. Insert O-rings if required.

Positioning and securing the clamping fixture

8. Carefully position the clamping fixture (01) on the spindle nose of the workpiece spindle (07), ensuring that it is aligned correctly. Observe the position of the positioning bore (02) for the positioning pin (06), identified by the  symbol.

The positioning pin (06) must be located in the positioning bore (02) provided.

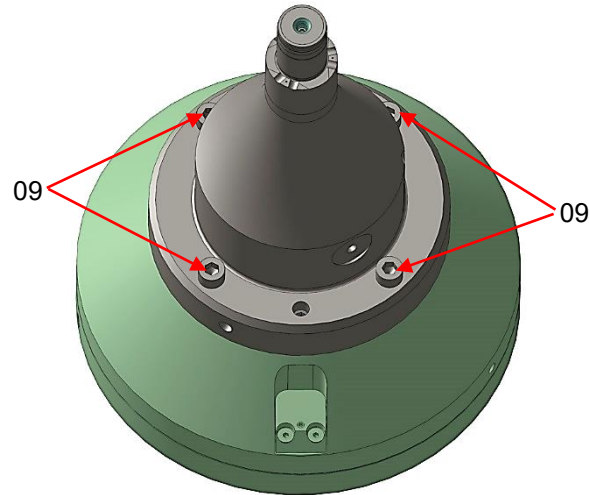


Fig. 15: Secure type C clamping fixture (top view)

9. Position four fastening screws (09) in the bores provided.
10. Slightly tighten four fixing screws (09) in a crosswise manner.
11. Tighten the fixing screws (09) in a crosswise manner to a torque of max. 20 Nm for M8 screws and 30 Nm for M10 screws.

Check the radial and axial runout

12. Check the radial and axial runout using a gauge with an indicating caliper: See drawing provided for the clamping fixture.



If the radial and axial runout deviations suggest a higher value than specified, remove the clamping fixture and repeat steps 4 to 11.

If the deviations are still not permissible, send the clamping fixture to the manufacturer (see section *Return Transportation to Manufacturer*) and fit a new clamping fixture.

13. Fit support ring and intermediate bush if required → *Fit/Remove Support Ring and Pendulum System* and *Fit/Remove Intermediate Bush* sections.
- > The type C clamping fixture is now fitted.

5.4 Fit/remove support ring and pendulum system

Prerequisites

- The clamping fixture is installed → section 5.3.
- The machine tool is prepared → section 5.2.

5.4.1 Fit support ring

Tools

- Lint-free cloth
- Allen wrench set
- Torque wrench with hexagon socket



In the example shown, the support ring is held with just three screws. The support ring can also be fixed using more screws.

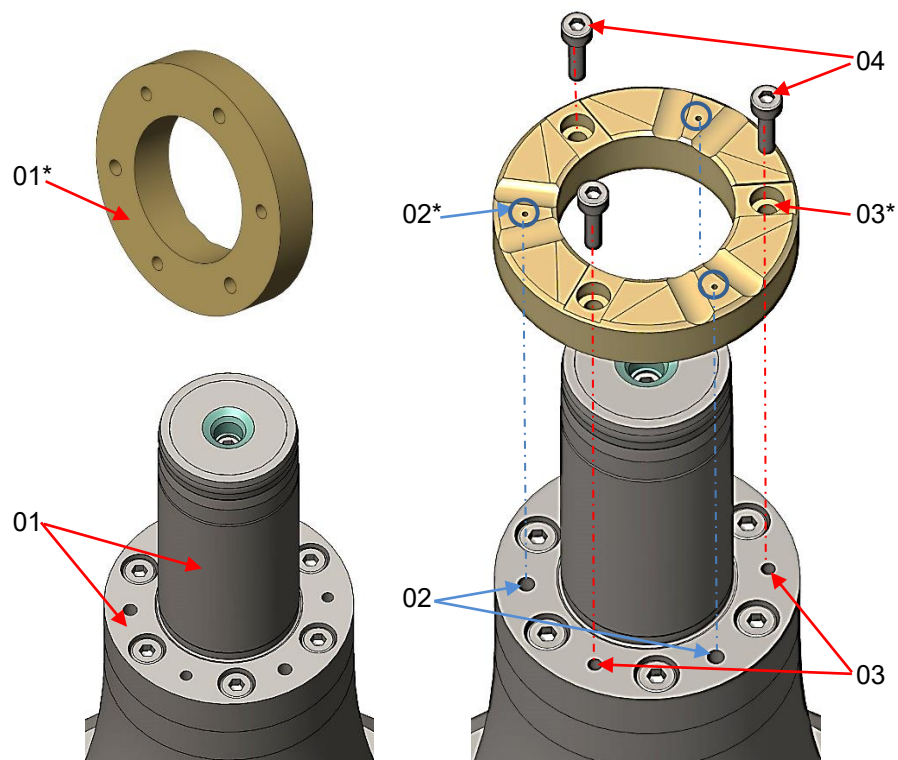


Fig. 16: Fit support ring

1. Clean the locating surfaces of the support ring (01*) and the clamping arbor (01) with a lint-free cloth and check for damage.
2. Place the support ring (01*) on the clamping fixture as follows:
 - The air holes (02*) for part present monitoring must be aligned with the air holes (02) on the clamping fixture.
 - The bores (03*) must be aligned with the bores (03) for the fixing screws (04).

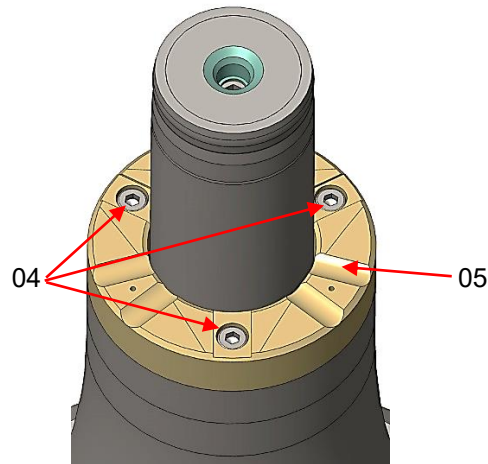


Fig. 17: Fit/remove support ring

3. Position the fixing screws (04) in the bores provided.
 4. Tighten the fixing screws (04) using an Allen wrench to the appropriate torque (see torque chart in the Appendix).
- > The support ring (05) is now fitted.

5.4.2 Remove support ring

1. Unscrew and remove fixing screws (04) using an Allen wrench.
 2. Remove the support ring (05) from the clamping fixture.
- > The support ring is now removed.

5.4.3 Fit pendulum system

Tools

- Lint-free cloth
- Screwdriver set

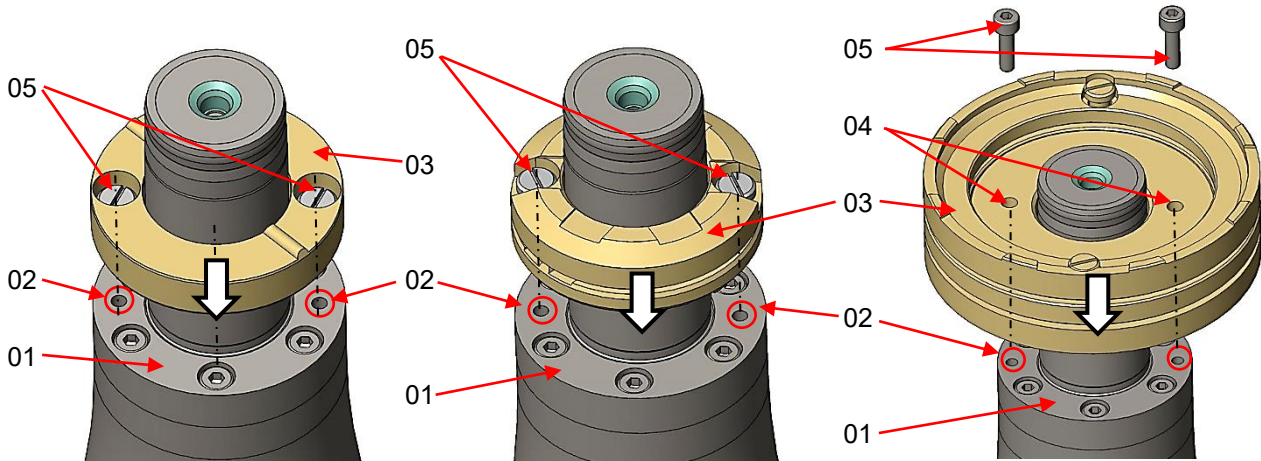


Fig. 18: Pendulum systems: one-piece, two-piece, three-piece

1. Clean the locating surfaces (01) of the clamping arbor and pendulum system (03) with a lint-free cloth and check for damage.
2. Place the pendulum system (03) on the clamping fixture in such a way (white arrow) that the bores (02) on the clamping arbor (marked in red) are aligned with the bores (04) for the fixing screws (05).

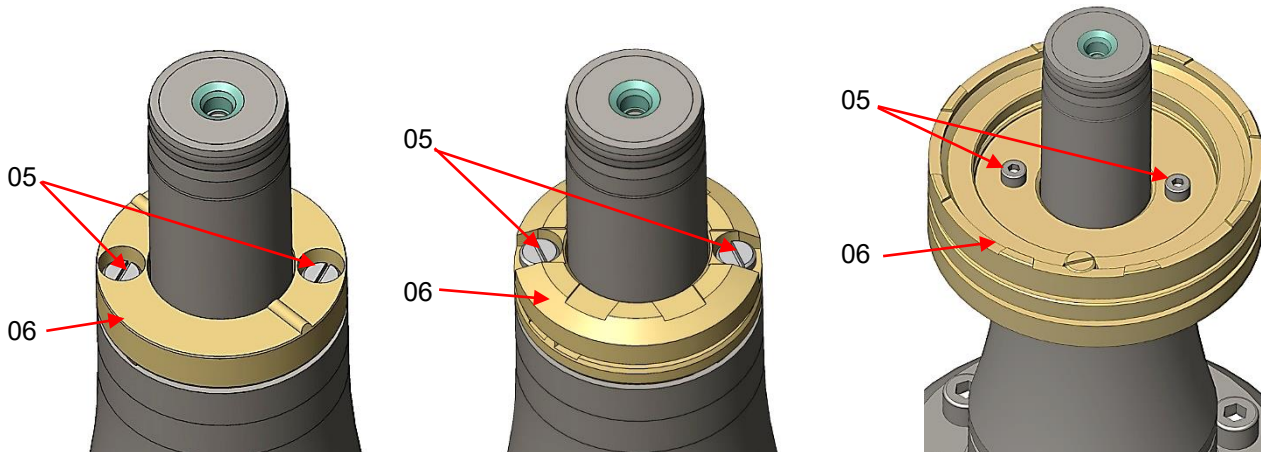


Fig. 19: Pendulum systems: one-piece, two-piece, three-piece

3. Position the fixing screws (05) in the bores provided.
 4. Tighten fixing screws (05) using a screwdriver in a clockwise direction.
- > The pendulum system is now fitted.

5.4.4 Remove pendulum system

1. Use a screwdriver to unscrew fixing screws (05) in a counterclockwise direction and remove.
 2. Remove the pendulum system (06) from the clamping fixture.
- > The pendulum system is now removed.

5.5 Fit/remove intermediate bush

Prerequisites

- The clamping fixture is installed → section 5.3.
- The machine tool is prepared → section 5.2.

5.5.1 Fit the intermediate bush using retaining washer and screws (variant 1)

Tools

- Lint-free cloth
- Torx wrench set

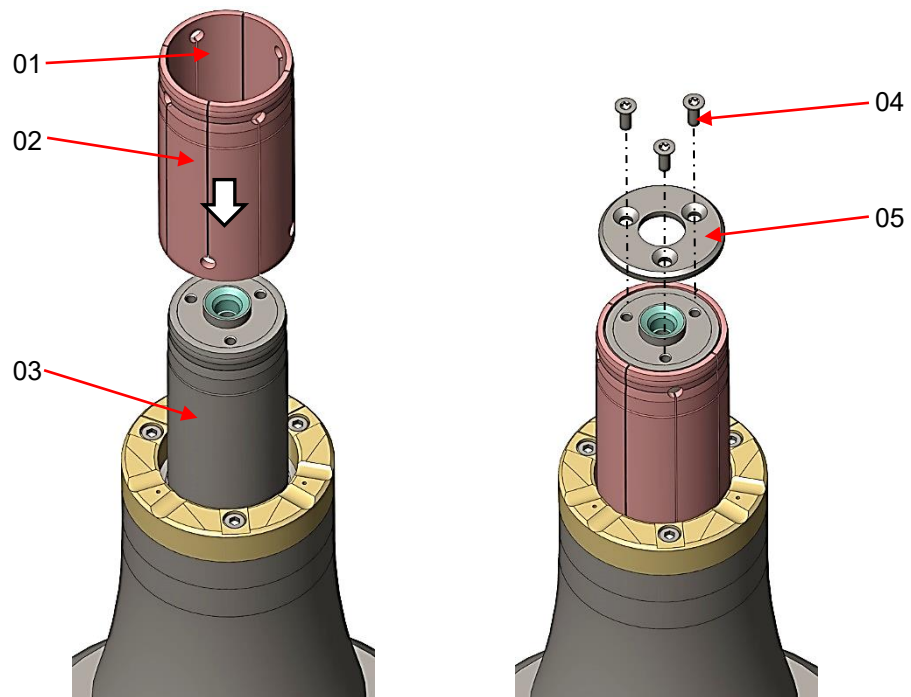


Fig. 20: Fix the retaining washer using three screws

1. Clean the locating surfaces of the clamping fixture (03) and the intermediate bush (01) with a lint-free cloth and check for damage.
 2. Slide the intermediate bush (02) over the clamping bush (03) (arrow).
 3. Position the retaining washer (05)
 4. Insert screws (04) into the designated bores and turn until hand-tight.
- > The intermediate bush is now fitted.

5.5.2 Remove intermediate bush (variant 1)

1. Unscrew fixing screws (04) using a Torx wrench and remove.
 2. Remove the retaining washer (05)
 3. Lift intermediate bush (02) upwards.
- > The intermediate bush is now removed.

5.5.3 Fit the intermediate bush using retaining washer and locking ring (variant 2)

Tools

- Lint-free cloth

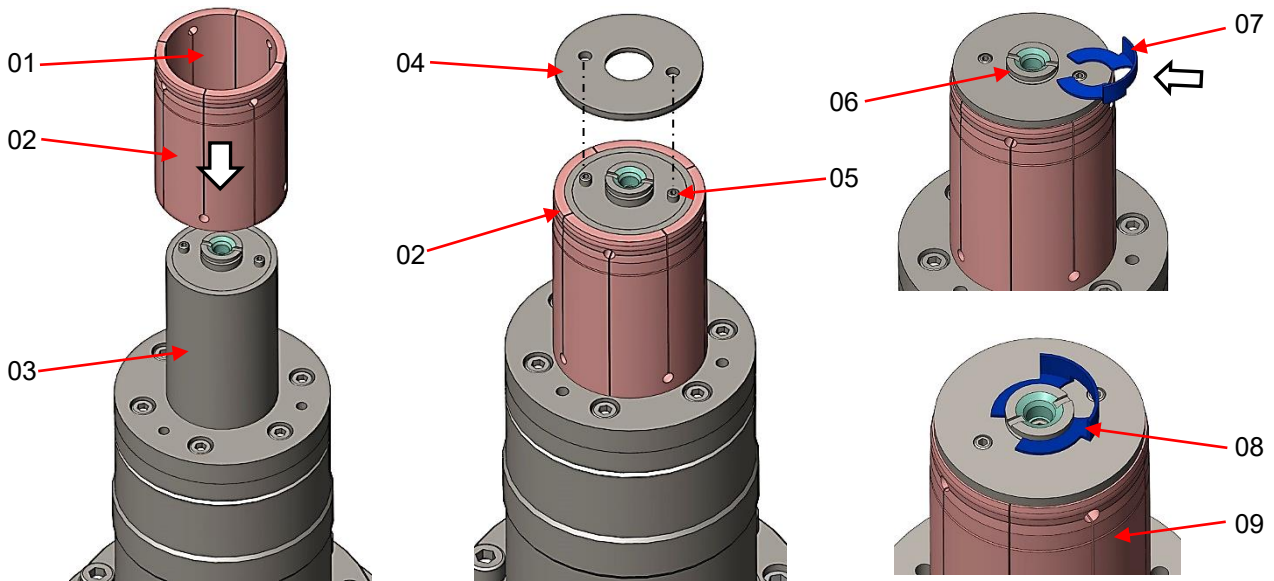


Fig. 21: Fix the retaining washer using locking ring.

1. Clean the locating surfaces of the clamping fixture (03) and the intermediate bush (01) with a lint-free cloth and check for damage.
2. Slide the intermediate bush (02) over the clamping bush (03) (arrow).
3. Place the retaining washer (04) on the intermediate bush (02) in such a way that the threaded pins (05) are located in the designated bores on the retaining washer (04).
4. Push the locking ring (07) sideways into the designated nut (06) (arrow) until the locking ring engages,
 - > The locking ring (08) and intermediate bush (09) are now fitted.

5.5.4 Remove intermediate bush (variant 2)

1. Unscrew locking ring (08) and remove.
2. Detach the retaining washer (04) and remove.
3. Lift intermediate bush (02) upwards.
 - > The intermediate bush is now removed.

5.5.5 Fit intermediate bush on support ring (variant 3)

Prerequisite

- The support ring is installed → section 5.4.2.

Tools

- Lint-free cloth
- Allen wrench set

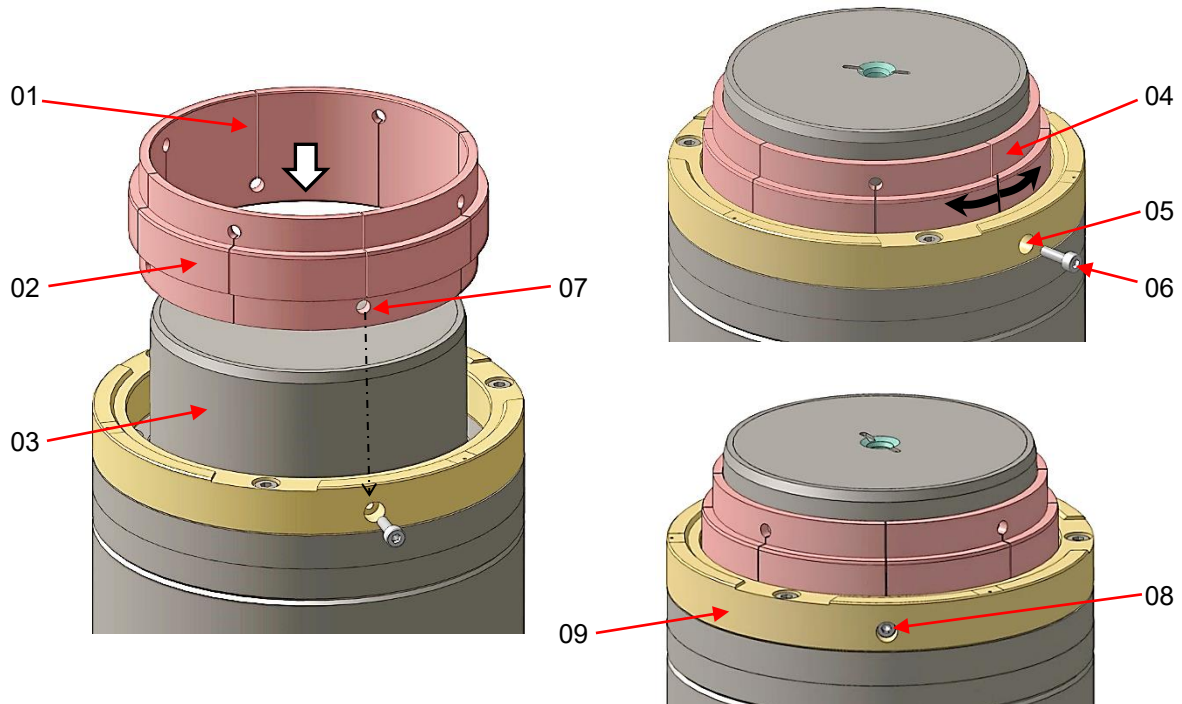


Fig. 22: Fix the intermediate bush on the support ring

1. Clean the locating surfaces of the clamping fixture (03) and the intermediate bush (01) with a lint-free cloth and check for damage.
2. Slide the intermediate bush (02) over the clamping bush (03), ensuring that it is aligned correctly (white arrow).
3. Rotate the intermediate bush (04) in such a way (black arrow) that the bores (07) of the intermediate bush are aligned with the bores (05) for the fixing screws (06) on the support ring.



In the example shown, the intermediate bush is held with just one radial screw (06). The intermediate bush can also be fixed using multiple screws.

4. Insert the fixing screw (06) into the designated bore (05) on the support ring.
 5. Ensure that the fixing screw (06) also enters into the bore (07) of the intermediate bush.
 6. Tighten the fixing screws (06 and/or 08) hand-tight using an Allen wrench.
- > The intermediate bush is now fitted.

5.5.6 Remove intermediate bush from support ring (variant 3)

1. Loosen the radial fixing screws (08) on the support ring (09) using an Allen wrench.
 2. Lift intermediate bush (04) upwards.
- > The intermediate bush is now removed.

5.6 Applying settings to the machine tool

1. Specify the clamping fixture in the data input under *Configuration – Specify clamping fixture* and enter the required values; see *Operating manual for gear grinding machine*.
2. Set and check clamping pressure, see *Operating manual for gear grinding machine*.

Incorrectly Set Clamping Pressure

WARNING



Risk of injury and material damage due to incorrectly set clamping pressure!

A clamping pressure that is too high can destroy the clamping fixture and cause serious injury. If the clamping pressure is set too low, the workpiece will not be sufficiently tensioned. This can lead to a collision with machine parts. Parts may be ejected, and the pressure medium may escape at high pressure. This can lead to serious or fatal injury as well as material damage.

- > Always set the clamping pressure at the clamping fixture according to the appropriate specification.
- > Never change the clamping pressure without authorization.
- > Ensure that the correct clamping pressure is set every time the clamping fixture is changed.

6 Operation

6.1 Operating safety

Personnel

- Technical personnel

Personal protective equipment

- Protective workwear
- Safety shoes
- Safety goggles if necessary
- Chemical-resistant protective gloves if necessary

WARNING



Risk of injury and material damage due to improper operation!

Improper operation can lead to serious injury or material damage and impair the functionality of the device.

- > Perform all operating steps in accordance with the information provided in these instructions.
- > Maintain order and cleanliness in the work area. Parts and tools that are not stacked securely or are lying around are a potential cause of accidents.
- > Do not loosen any additional screws on the device unless expressly instructed to do so in these instructions.

Breakage of the clamping bush

WARNING



Risk of injury and material damage due to improper tensioning of the clamping bush!

Tensioning the clamping bush without a workpiece or protection sleeve can result in destruction of the clamping bush and serious injuries.

- > Never tension the clamping bush without introducing a workpiece or protection sleeve.
- > Ensure that surface to be clamped is fully cylindrical and does not have any interruptions (i.e. grooves, bores, threads etc.).

Working with open protection doors in manual mode**WARNING**

Danger of injury due to moving machine components with open protection doors! A series of setup tasks are performed with the protection doors open. Necessary machine motions are rendered safe by the confirmation function. Even so, there is still a danger of injury in principle.

- > Perform the tasks with the protection doors open with extreme care and caution.
- > Ensure that tasks with open protection doors are only performed by authorized operating staff.
- > Do not reach into or linger near moving components.
- > The machine may only be operated by one person at a time.

6.2 Tasks during operation

The tasks of the operating personnel during operation are restricted to changing workpieces during semi-automatic operation and responding to control system alarms.

The PLC alarms are listed in the grinding machine operating instructions. For possible faults in the clamping fixture and information about fault elimination, please refer to the *Potential Problems and Corrective Actions* section.

Clamping stroke control (option)

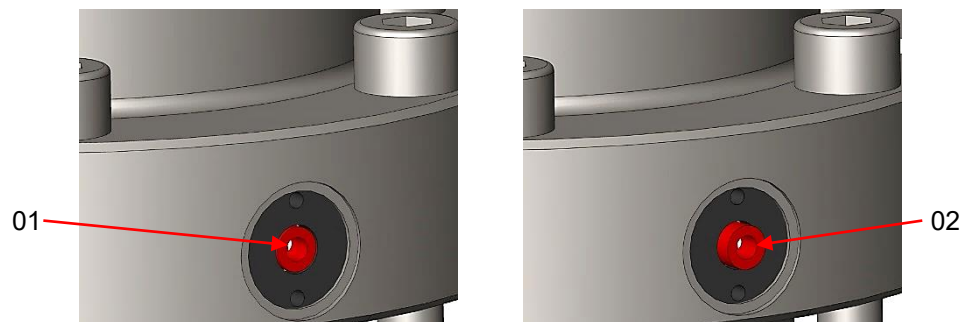


Fig. 23: Control pin of the clamping stroke monitoring

01	Control pin retracted – normal operation	02	Control pin extended – clamping device is not fully functional
----	--	----	--

The functionality of the clamping fixture can be checked using the visual clamping stroke control, provided that the function of the clamping stroke control is activated in the machine control.

If the fitted clamping fixture is no longer sufficiently functional due to loss of oil, the red control pin (02) protrudes by approx. 2 mm. At the same time, air escapes from the vent hole in the control pin. In this case, the clamping fixture must be replaced and sent for repair.

7 Decommissioning

7.1 Safety for decommissioning

Personnel

- Technical personnel

Personal protective equipment

- Protective workwear
- Safety shoes
- Safety goggles if necessary
- Chemical-resistant protective gloves if necessary

WARNING



Risk of injury and material damage due to improper decommissioning!

Improper decommissioning can lead to serious injury or material damage and impair the functionality of the device.

- > Perform all operating steps in accordance with the information provided in these instructions.
- > Maintain order and cleanliness in the work area. Parts and tools that are not stacked securely or are lying around are a potential cause of accidents.
- > Do not loosen any additional screws on the device unless expressly instructed to do so in these instructions.

7.2 Remove clamping fixture

7.2.1 Remove type A clamping fixture

Tools

- Allen wrench set

Preparations

1. Prepare machine tool → section 5.2.

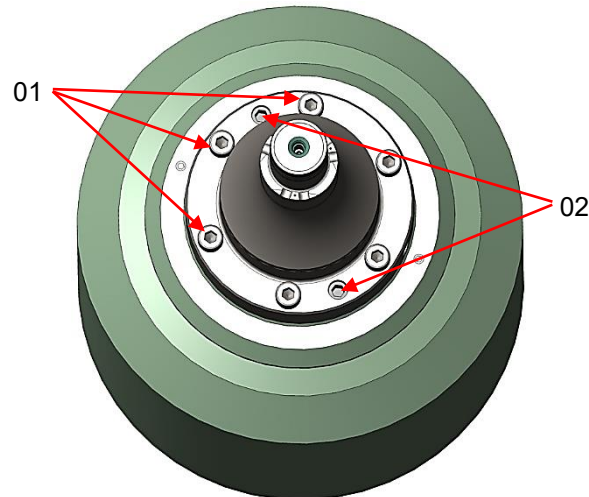


Fig. 24: Remove type A clamping fixture (top view)

2. Use an Allen wrench to unscrew six fixing screws (01) in a counterclockwise direction and remove.
 3. Screw in threaded lift-off pins (02) clockwise until the clamping fixture is loosened from the workpiece spindle.
 4. Slightly lift the clamping fixture and carefully remove from the workpiece spindle.
 5. Adjust threaded lift-off pins (02) on the clamping fixture anti-clockwise until they no longer protrude.
 6. Place the clamping fixture in the transportation container.
- > The clamping fixture is now removed.

7.2.2 Remove type B clamping fixture

Preparations

1. Prepare machine tool → section 5.2.
2. Remove the type B clamping fixture from the Base Unit → instructions provided for the relevant Base Unit.

7.2.3 Remove type C clamping fixture

Tools

- Allen wrench set

Preparations

1. Prepare machine tool → section 5.2.

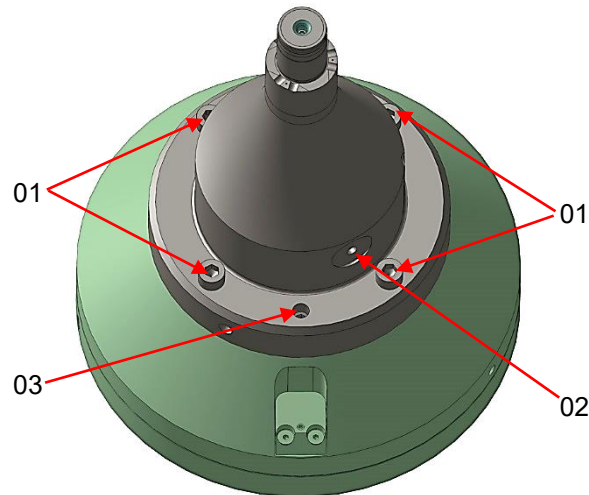


Fig. 25: Remove type C clamping fixture

2. Use an Allen wrench to unscrew four fastening screws (01) in a counterclockwise direction, and remove.
 3. Screw in threaded lift-off pins clockwise on the clamping fixture until the clamping fixture is loosened from the workpiece spindle.
 4. If required, screw the eye bolts into two lifting threads (02) of the clamping fixture.
 5. If required, attach cables or straps with sufficient load capacity to a crane.
 6. Slightly lift the clamping fixture and carefully remove from the workpiece spindle.
 7. Adjust threaded lift-off pins (03) on both sides by turning anti-clockwise until they no longer protrude.
 8. Place the clamping fixture in the Reishauer transportation container.
- > The clamping fixture is now removed.

8 Potential Problems and Corrective Actions

WARNING



Risk of injury and material damage due to improper repair work!

The device is not designed to be repaired by the operator. Improper repair work can lead to serious injury or material damage.

- > Never attempt to open or repair the device.
- > In the event of problems that cannot be rectified by the following information, remove the device and send to the manufacturer for repair → Section *Transportation, Packaging, and Storage* → *Return transportation to manufacturer*.

8.1 Faults table

Problem description	Possible cause	Remedy	Carried out by
Radial and axial runout on the fitted clamping fixture are incorrect	Loose or incorrectly fitted fixing screws	Remove clamping device and secure correctly.	Technical personnel
	Contaminated locating surfaces	Remove the clamping fixture and check the locating surfaces for contamination. If necessary, clean locating surfaces with a lint-free cloth.	Technical personnel
	Damaged locating surfaces	Remove the clamping fixture and check the locating surfaces for damage. If necessary, replace the clamping fixture. Send damaged clamping fixture for repair.	Technical personnel Manufacturer (repair)
	Protruding threaded lift-off pin	Remove the clamping fixture, adjust threaded lift-off pin, and refit the clamping fixture.	Technical personnel
	Oil leaks or mechanical wear at the clamping location	Check the control pin of the clamping stroke control on the clamping fixture or Base Unit (option). If the red control pin protrudes 2 mm, replace the clamping fixture.	Technical personnel
Clamping fixture does not fit into the holder on the base unit	Incorrect connecting dimensions of the clamping fixture	Check the connecting dimensions of the clamping fixture. If necessary, replace the clamping fixture.	Technical personnel
	Damaged locating surfaces	Check the locating surfaces for damage. If necessary, exchange the base unit or clamping fixture.	Technical personnel
Clamping fixture does not tension or the clamping force is too low	Reduced clamping stroke of the clamping piston due to foreign objects or contaminated mechanics	Check operating piston and intermediate pieces in the workpiece spindle, Base Unit or clamping fixture for contamination and foreign objects.	Technical personnel
	Incorrect axial position of the workpiece	Check whether the workpiece is in the clamping position.	Technical personnel

Problem description	Possible cause	Remedy	Carried out by
	Loss of oil in the clamping fixture to be tensioned	Check the control pin of the clamping stroke control. If the red control pin protrudes 2 mm, replace the clamping fixture (option).	Technical personnel
		Check whether air emerges through the vent hole of the clamping stroke control (option) during the clamping procedure or if a fault message is pending; if necessary, replace the fitted clamp fixture.	Technical personnel
		Check whether the piston in the clamping fixture can be manually pressed without hindrance. In this case, replace the clamping fixture.	Technical personnel
Part present monitoring on the clamping fixture to be tensioned no longer works	Hydraulic or grinding oil in the pneumatic line	Activate blow-out function and clean pneumatic line with blow-out air.	Technical personnel
	Axial backstop is in incorrect position or is missing	Check position of the support rings at the air connections.	Technical personnel
	Pneumatic unit switched off	Switch on pneumatic unit	Technical personnel

9 Transportation, Packaging, and Storage

9.1 Transportation safety

WARNING



Danger of death due to suspended loads!

There is a danger of death when lifting loads due to falling or uncontrolled suspended parts.

- > Never step under suspended loads.
- > Observe the information regarding the attachment points provided (see Section 9.3 *Transportation*).
- > Only use the permitted lifting devices and slinging gear with sufficient load-bearing capacity. Make sure the slinging gear is securely fastened.
- > Do not use any torn or worn out cables and belts.

Improper transportation

ATTENTION

Damage caused by improper transportation!

In the event of improper transportation, transported items may fall or be knocked over. This may result in significant material damage.

- > Proceed carefully when unloading consignments upon delivery and during internal transportation.
- > Only use the attachment points provided.
- > Remove packaging only shortly before commissioning.
- > Observe the storage conditions when returning goods to the manufacturer → Section 9.6 *Storage*.

9.2 Transportation conditions

Specification	Value
Temperature range	-5 to 40 °C
Relative humidity, max.	20 to 70 %

9.3 Transportation

Attachment points

Lifting threads are provided on heavy clamping fixtures (15 kg or more) → *Construction and Function* section and attached drawing. The lifting threads are used for lifting the clamping fixture with the customer's own lifting device.

9.4 Packaging

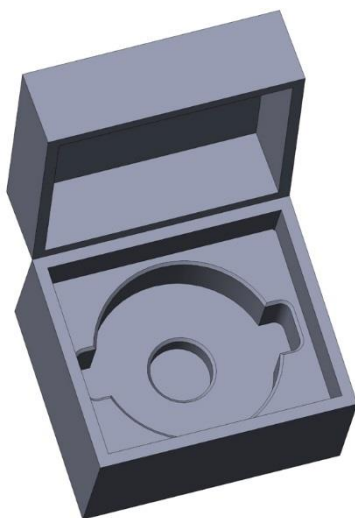


Fig. 26: Interior layout of Reishauer transportation container

The device is packaged in a Reishauer transportation container in accordance with the expected transportation conditions.

This special packaging should protect the individual components against transportation damage, corrosion, and other damage until assembly. Therefore, do not destroy the packaging and only remove shortly before assembly. Retain the packaging in the event that the device must be returned to the manufacturer.

Observe the following if the packaging is to be disposed of:

ATTENTION

Environmental damage due to improper disposal!

Disposing of the packaging incorrectly can cause significant damage to the environment.

- > Dispose of the packaging materials in an environmentally friendly manner.
- > Observe disposal regulations applicable locally.

9.5 Return transportation to manufacturer

1. Remove the device.
2. Pack the device in its original packaging (Reishauer transportation container).
3. The red accompanying document "Clamping device for repair/corrective work" must be filled in.

REISHAUER action sheet – clamping device

Ship repair / rework to:	Sender / Return address:
Reishauer AG	Address:.....
Industriestrasse 40	
CH-8304 Wallisellen	
	Contact name:.....
	
Reishauer description	Customer description
Item number:	Tool number:
Serial number:	Gear description/name:
Machine parameters	
Clamping pressure:bar	clamping force of tail stock:.....N rpm of C-Axis:min ⁻¹
Tool life	
Number of clamping cycles:	

Fig. 27: Part of the accompanying document "Clamping device for repair/corrective work"



The red accompanying document can be found in the Reishauer transportation container and contains information that will be required to carry out repair or corrective work on the clamping device (see *Accompanying document* in the Appendix). Orders without this information cannot be considered.

4. Send to the manufacturer. Please see the last page of these instructions for the manufacturer's address.

9.6 Storage

ATTENTION

Material damage due to improper storage!

Improper storage can result in a high level of material damage.

> Always observe the information listed below regarding storage.

Preparing device for storage

Personnel

- Technical personnel

Personal protective equipment

- Protective workwear
- Safety shoes
- Safety goggles
- Chemical-resistant protective gloves

1. Remove the device.
2. Clean device with a lint-free cloth.
3. Lightly lubricate the device with acid-free oil.
4. Pack the device in its original packaging.

Always store the device under the following conditions

- Never store outside.
- Store in a dry and dust-free place.
- Do not expose to aggressive media.
- Protect from sunlight.
- Avoid mechanical vibrations.
- Storage temperature: 15 to 45°C.
- Relative humidity: max. 70%.
- Regularly check the condition of all parts and the packaging when storing for longer than 3 months. If necessary, renew or replace conservation agent.

9.7 Disposal

Once the device has reached the end of its useful life, the device must be removed from the machine and sent to the manufacturer for disposal. Please see the last page of these instructions for the manufacturer's address.

10 Appendix

10.1 Drawing labels – Clamping arbor, hydraulic DH

Wording on drawing (German)	Wording on drawing (English)	Translation	Value
Maschinendruck	Machine pressure		-
Spannhubüberwachung	clamping stroke control		-
Hydraulikanschluss P	Hydraulic connection P		-
Prüfbund	reference diameter		-
Beschriftung	marking		-
Spannhub/Hub	Clamping stroke/strok		-

10.2 Torque table

The table lists the screw-tightening torque required to attain the maximum pre-tension for metric standard threads, in Nm. The screws are greased.

Diameter	Strength class
	12.9
M4	4
M5	8
M6	13
M8	32
M10	62
M12	108

10.3 Job ticket for repair/rework of clamping device

Ship repair / rework to:

Sender / Return address:

**Reishauer AG
Industriestrasse 40
CH-8304 Wallisellen**

Address:.....

.....

.....

.....

.....

Abt. 3000

Contact name:

.....

Reishauer-description:

Item number:

Serial number:

Customer description:

Tool number:

Gear description/name:

.....

Machine parameters:

Clamping pressure:bar clamping force of tail stock:N rpm of C-Axis:min⁻¹

Tool life: Number of clamping cycles:

Documentation and description of objection

Diagram of last workpiece: ☐ as attachment

Objection:

☐ workpiece can not be loaded

☐ workpiece does not clamp

☐ concentric runout not o.k.

☐ axial runout not o.k.

☐ Expansion is too low

☐ clamping bush is worn

☐ clamping bush is scratched

☐ clamping device is leaking

☐ base is worn

☐ collision

☐ see remark

Remark:

Gear specifications:

Drawing and gear specifications for workpiece: ☐ as attachment / ☐ no attachment

Attention !

**To control the clamping device the control ring is necessary
Please provide the corresponding control ring for each clamping device**

Attachments:

.....

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REISHAUER

Gear Grinding Technology